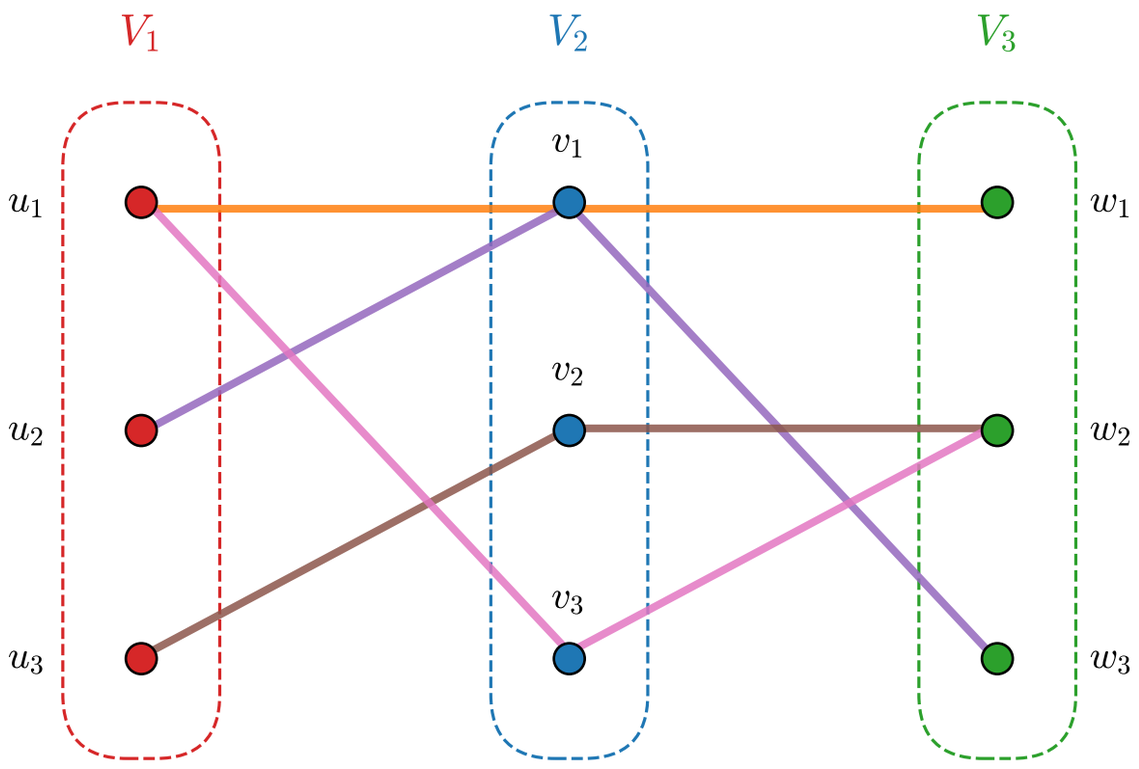
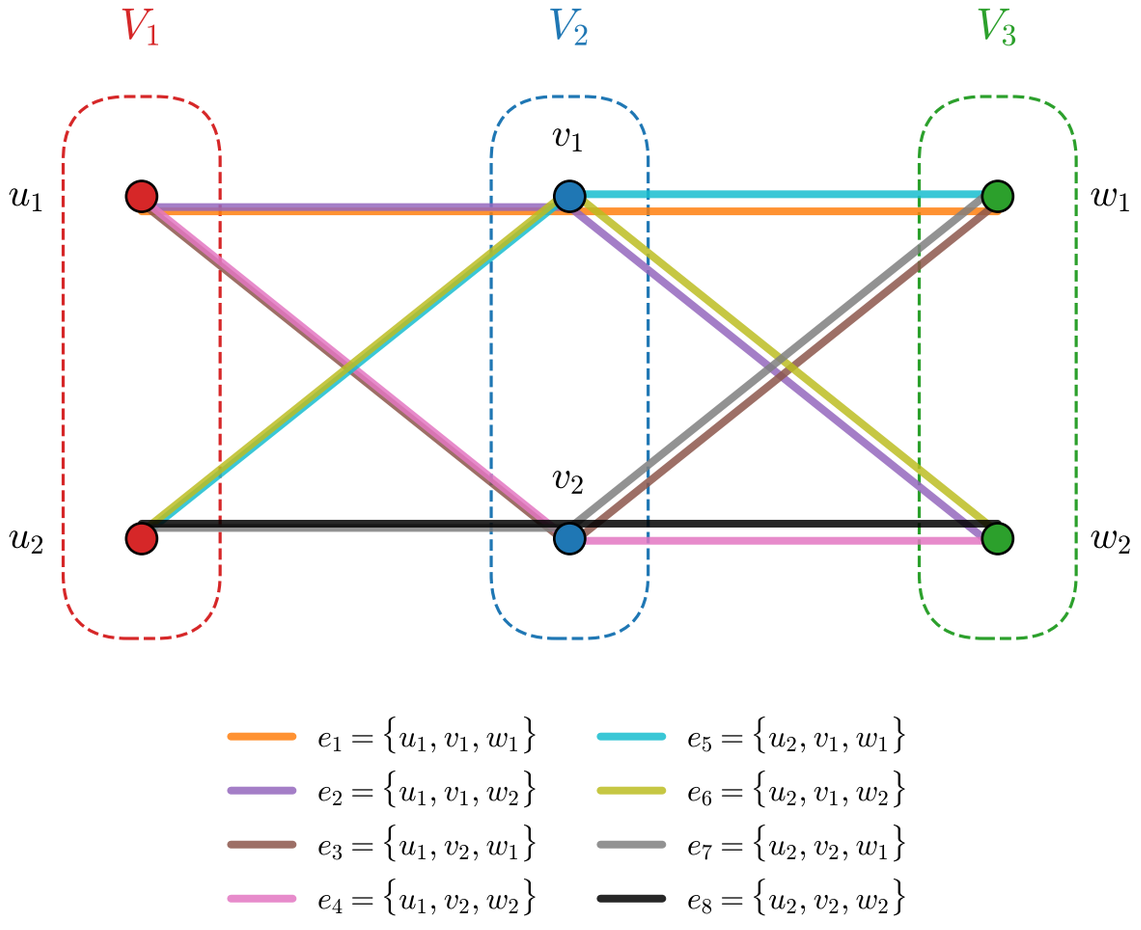
(a) 3-partite 3-uniform: one vertex from each part



(b) Complete $K^{(3)}_{2,2,2}$: $2 \cdot 2 \cdot 2 = 8$ edges